

Mall Customer Segmentation Using K-Means Clustering

Project Introduction

This project focuses on customer segmentation using the K-Means Clustering algorithm. The dataset contains customer information such as age, gender, annual income, and spending score. The objective of this project is to divide customers into different groups based on their purchasing behavior.

Dataset Features

- CustomerID – Unique customer ID
- Gender – Male/Female
- Age – Customer age
- Annual Income (k\$) – Annual income of customer
- Spending Score (1-100) – Spending behavior score assigned by mall

Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn

Machine Learning Algorithm

K-Means Clustering is an unsupervised machine learning algorithm used to group similar customers into clusters based on income and spending patterns.

Project Workflow

1. Import libraries
2. Load dataset
3. Perform data visualization
4. Use Elbow Method to find optimal clusters
5. Apply K-Means Clustering
6. Visualize customer segments
7. Add cluster labels to dataset

Cluster	Description
Cluster 1	High income, high spending customers
Cluster 2	High income, low spending customers
Cluster 3	Average customers
Cluster 4	Low income, high spending customers
Cluster 5	Low income, low spending customers

Conclusion

The project successfully segmented mall customers into different groups using K-Means Clustering. These customer segments help businesses improve marketing strategies, customer targeting, and personalized recommendations.